RGBT Fusion-Based Airborne Synthetic Aperture Occlusion Removal Target Imaging

Yuan Yue, Chunna Tian, *Member, IEEE*, Zhiwei Jiang, Xiangyang Li, Shiguo Chen, and Dongliang Hou

***Abstract*—Drones are gaining increasing popularity in bush search and rescue operations, where enhancing the accuracy of occluded target detection is crucial to find the targets. However, airborne light detection and ranging (LiDAR) and multispectral cameras are limited by cost, efficiency, and imaging quality, while RGB- or thermal-based detection methods rely heavily on the model performance and prior knowledge. Recovering the target in imaging is a new frontier for occluded target perception. Here, we propose a novel synthetic aperture fusion imaging (SAFI) method for drones, combining an RGBT fusion network with synthetic aperture imaging (SAI) to recover occluded targets. First, we propose a color-contrast RGBT fusion network (C-RGBTNet) based on a transformer. C-RGBTNet captures target thermal features from thermal images and assigns RGB color features to occlusions, generating high-quality fusion images. Then, we apply hue, saturation, and value (HSV)-based color segmentation to isolate the target with strong thermal intensity (high brightness) from the cluttered background. Finally, we perform SAI, synthesizing the different segmented target parts to yield an occlusion-free target image. Experimental results in our home-brew downward-view multimodal bush (DMB) dataset demonstrate that SAFI achieves high-quality target imaging with effective occlusion removal, significantly outperforming traditional methods.**

***Index Terms*—Airborne synthetic aperture imaging (ASAI), color segmentation, image fusion, occlusion removal target imaging.**

I. Introduction

RECENT advancements in drone technology have expanded their applications, particularly in bush search and rescue operations [1], [2], [3]. Equipped with light detection and ranging (LiDAR) and multispectral cameras, drones can collect multimodal data to support target detection across complex environments. LiDAR offers accurate 3-D point clouds but with limited efficiency, while multispectral 2-D/3-D imaging enables rapid detection. However, occlusion often hinders the ability of multispectral cameras to capture complete target information.

To address occlusion, research has primarily focused on algorithmic improvements. Traditional algorithms depend on manually designed features and prior knowledge, such as optical flow [4], background subtraction [5], and object tracking with occlusion detection [6]. Deep-learning-based [7] detection algorithms like YOLO [8] and Mask R-CNN [9] use a learnable convolutional neural network (CNN) to extract and identify features, which improves performance under occlusion. In addition, transformer-based [10] methods use attention mechanisms to obtain global target information, further enhancing the robustness and accuracy of occluded target detection. However, their effectiveness is heavily reliant on accurate modeling and prediction with prior information about the target, limiting their generalizability and reliability. From the imaging perspective, Faraday [11] first proposed that "light should be interpreted as a field." With advancements in computers and digital cameras, the light field imaging has transitioned from theoretical exploration to practical applications. By fusing local images from multiple perspectives and adjusting the focal plane, light field imaging can perform perspective imaging of partially occluded targets [12], [13]. However, existing light field cameras face challenges in resolution, size, weight, and power consumption, which hinder their integration into drones, leaving light field imaging promising yet underexplored in drone applications.

Airborne synthetic aperture imaging (ASAI) [14], [15] is a novel technique designed to address the challenges faced by drones in occluded scenarios during search and rescue [16], [17]. ASAI employs a low-cost, small-aperture camera mounted on a drone to capture downward-view images from multiple positions, simulating a camera array. The images are synthesized into a large-aperture image, which is then refocused to aggregate exposed information of the target for occlusion-free imaging. ASAI relies solely on images without 3-D reconstruction, allowing for low computational cost and rapid imaging. However, applying synthetic aperture imaging (SAI) [18] to thermal images struggles in heavily occluded scenarios, where blurred occlusions obscure the target. To solve this problem, the color anomaly detection is incorporated before SAI, known as synthetic aperture anomaly imaging (SAAI) [19]. SAAI first detects all abnormal pixels in the thermal images and then performs SAI on the results. The salient thermal features of the target allow the refocused large-aperture image to contain only the target. Nevertheless, the

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Yuan Yue, Chunna Tian, and Xiangyang Li are with the School of Electronic Engineering, Xidian University, Xi'an 710071, China (e-mail: yueyuan@stu.xidian.edu.cn; chnatian@xidian.edu.cn; lxy@stu.xidian.edu.cn).

Zhiwei Jiang and Shiguo Chen are with the School of Physics, Xidian University, Xi'an 710071, China (e-mail: 23201110762@stu.xidian.edu.cn; sgchen@xidian.edu.cn).

Dongliang Hou is with the School of Telecommunications Engineering, Xidian University, Xi'an 710126, China (e-mail: 22009100103@stu.xidian.edu.cn).

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traditional color anomaly detection often confuses occlusions with thermal intensity similar to the target and also incorrectly identifies low-thermal occlusions as anomalies. This leads to numerous pixel clusters in the final imaging result, some of which even adhere to the target, making subsequent detection more difficult.

Recently, deep multimodal fusion (e.g., RGBT) has shown obvious advantages over single-modality approaches in various applications [20], [21]. Thermal images offer thermal information under occluded conditions, while RGB images provide rich texture. Their integration enhances the identification of obscured or hidden targets. The transformer-based encoder–decoder model, proposed by Vaswani et al. [10], has demonstrated remarkable performance improvements in the image fusion compared with CNN-based methods [22], [23], [24]. To this end, we propose a color-contrast RGBT fusion network (C-RGBTNet) based on a transformer to address the challenge of insufficient occlusion removal caused by the difficulty in distinguishing targets from occlusions. C-RGBTNet offers global modeling capabilities through self-attention (SA), allowing it to capture long-range dependencies dynamically and adaptively focus on informative regions across both modalities. In addition, the cross-attention module enables the comprehensive integration of features from thermal and RGB images by facilitating interaction across all feature dimensions. Consequently, C-RGBTNet can generate fusion images that highlight thermal targets, preserve more details, and exhibit richer and more natural color information. In addition, most existing downward-view drone datasets are single modality, and typically face challenges such as minimal target occlusion or limited seasonal variation. Therefore, we create a dataset comprising both thermal and RGB images of the bush under varying levels of occlusion across three seasons, offering a more comprehensive resource for diverse experimental needs.

In this article, we propose synthetic aperture fusion imaging (SAFI) to address the limitations of SAAI. SAFI replaces the color anomaly detection with a hue, saturation, and value (HSV)-based color segmentation and incorporates an RGBT fusion network to achieve high-quality occlusion-removal target imaging, as shown in Fig. 1. Overall, the main contributions are as follows.

1) We propose a C-RGBTNet based on a transformer that integrates the RGBT complementary information, preserving salient thermal features of targets and RGB color features of occlusions effectively.
2) Leveraging the distinct color difference between targets and occlusions in the fusion image, we employ an HSV-based color segmentation to effectively extract the salient features of the target before applying SAI. The target in the fusion image exhibits high brightness, making the HSV color space especially effective for processing.
3) We construct a downward-view multimodal bush (DMB) dataset for an occluded target (a person lying down) across three seasons. The dataset includes slightly and heavily occluded scenarios. Within our dataset, SAFI demonstrates significant improvement over SAAI and SAI, achieving high-quality occlusion removal target imaging in all scenarios.

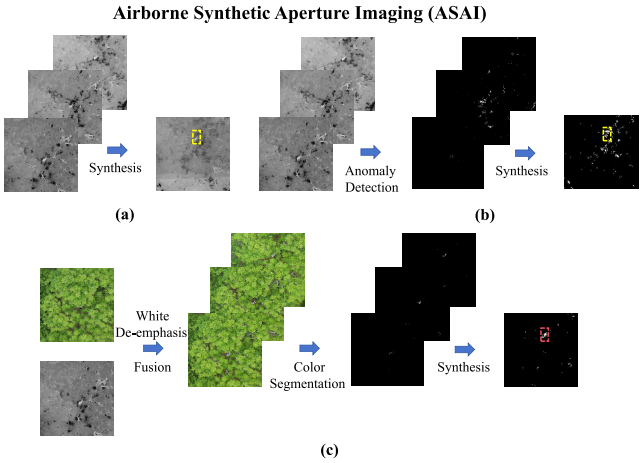


Fig. 1. Comparisons between three ASAI methods. (a) SAI. (b) SAAI. (c) Our SAFI. The red dotted box in (c) indicates the target (a person lying down) after occlusion removal, while the yellow dotted boxes in (a) and (b) represent the lost target or poor imaging results.

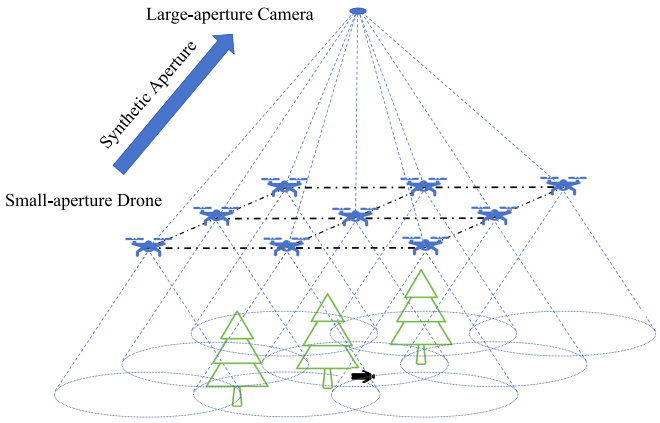


Fig. 2. Method for simulating a large-aperture camera using a small-aperture thermal camera on a drone to form a camera array while flying at different positions over a bush where a person is lying down.

II. Related Works

A. Airborne Synthetic Aperture Imaging

In dense bush, heavily occluded targets (e.g., humans and animals) are barely invisible in RGB images or unidentifiable in single thermal images. However, they can be recovered by integrating multimodal data from different views, particularly leveraging thermal information [18], [25], [26]. As shown in Fig. 2, ASAI uses a drone equipped with a small-aperture camera to capture downward-view images from multiple positions while flying, forming a camera array [15]. The synthesized array simulates a large-aperture camera, enabling high-resolution imaging without bulky light field cameras. Table I summarizes all ASAI methods, both SAI [16] and SAAI [19] process thermal images; however, their imaging quality is adversely affected by occlusions. To address these issues, we propose SAFI, which fuses thermal and RGB

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TABLE I
SUMMARY OF ASAI

Methods	Publication	Core Ideas	Performance
SAI [18]	RS	Synthesizing multi-view thermal images to refocus target	blurred occlusions obscure the target
SAAI [19]	RS	Synthesizing the RX [27] detected color anomalies in thermal images	occlusions are falsely detected as anomalies
SAFI (ours)	-	Synthesizing the HSV segmented results in fusion images	Effectively remove the occlusions

TABLE II
SUMMARY OF REPRESENTATIVE ENCODER-BASED RGBT IMAGE FUSION METHODS

Methods	Publication	Aspects	Performance	Efficiency(ms)
U2Fusion [22]	TPAMI	CNN	The intensity of the target pixel is insufficient	2914
CDDFuse [23]	CVPR	Transformer	The texture and brightness appear unnatural	1014
PromptFusion [24]	JAS	Transformer	High-thermal occlusions lack complete RGB color features assignment	962.6
C-RGBTNet (ours)	-	Transformer	Effectively distinguish between the occlusions and the target	161.9

The efficiency evaluation is performed on an NVIDIA GeForce RTX 3090 GPU with 10 randomly selected images from the DMB dataset.

images to preserve salient thermal features of targets and RGB color features of occlusions. The targets are characterized by high brightness (i.e., pure white) in the fusion image, enabling HSV segmentation for extraction, followed by SAI on the segmented results. SAFI addresses the issue of Reed–Xiaoli (RX) [27] false or missed detections caused by the confusion between targets and occlusions in SAAI, enabling high-quality target imaging in various occluded scenarios.

B. Fusion of RGB and Thermal Images

RGB and thermal image fusion aim to integrate complementary information from both modalities to generate fused images with richer details. Deep-learning-based RGBT image fusion methods are typically categorized into generative- and encoder-based models. Generative models learn the statistical distribution of a dataset and generate new data instances by sampling from this distribution, such as generative adversarial networks (GANs) and diffusion models. Rao et al. [28] introduced AT-GAN, which integrates intensity attention and semantic transition modules to extract key thermal and RGB information. Zhao et al. [29] proposed a denoising diffusion model-based fusion algorithm that formulates fusion as a conditional generation problem. Unlike generative models, encoder-based models map input data into a compact representation space to extract key information directly. Han et al. [22] proposed U2Fusion, an unsupervised CNN-based method that jointly learns shared and specific representations for image fusion. Zhao et al. [23] introduced CDDFuse, a dual-branch Transformer–CNN framework that decomposes global and local features for high-quality fusion. Liu et al. [24] propose PromptFusion, a framework that integrates semantic prompts into the fusion network to exploit vision–language constraints, improving fusion quality and downstream task performance. With advances in RGBT image fusion, fusion images provide salient features for tasks such as object detection, segmentation, and low-light enhancement. Since generative models are more time-consuming, we propose C-RGBTNet based on the transformer to separate targets and occlusions efficiently. Table II summarizes the representative encoder-based RGBT image fusion methods. Compared with other methods, C-RGBTNet enhances color distinction between targets and occlusions in fusion images by extracting prominent thermal and RGB features, effectively avoiding confusion. Our method also achieves a much lower average runtime on the DMB dataset. With the emphasis of SAFI on performance and efficiency, C-RGBTNet is the optimal method.

III. METHODS

The overall architecture of the proposed SAFI is illustrated in Fig. 3. Initially, a drone captures downward-view images of the scenario, including both thermal and RGB images. These images are then processed in five stages. First, thermal and RGB images are registered to align their resolution. Second, camera poses are estimated using RGB images. To mitigate the impact of near-white or pure-white occlusions on HSV color segmentation, white de-emphasis is applied to all RGB images before fusion. Third, the processed thermal and RGB images are fed into the fusion network (C-RGBTNet), which generates a fusion image that preserves the prominent features from both thermal and RGB images. Fourth, HSV color segmentation is applied to extract high brightness and white-colored thermal information from the target. Fifth, the segmentation results are mapped to the corresponding camera poses for SAI, which yields an occlusion-free target image.

A. Transformer-Based C-RGBTNet

In SAI, a heavily occluded target in RGB images is barely visible, while it is notably impacted by occlusions with thermal crossing in thermal images. To avoid confusion between targets and occlusions, we propose C-RGBTNet, which integrates prominent thermal and RGB features through a Transformer-based fusion network to enhance color distinction and achieve target–occlusion separation. As illustrated in Fig. 4, C-RGBTNet contains two main parts. First, a dual-branch transformer encoder to extract discriminatively fusion

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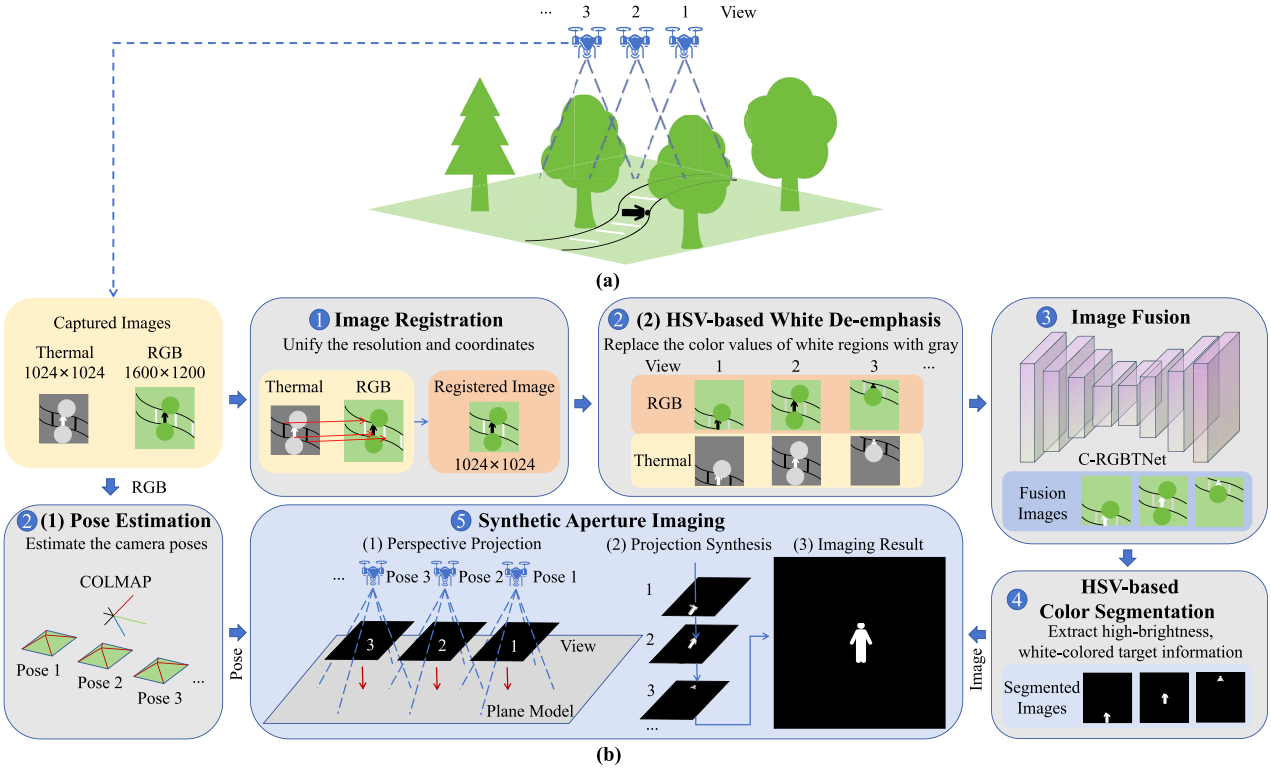


Fig. 3. Overall architecture of our proposed SAFI. (a) Image acquisition. (b) Image processing.

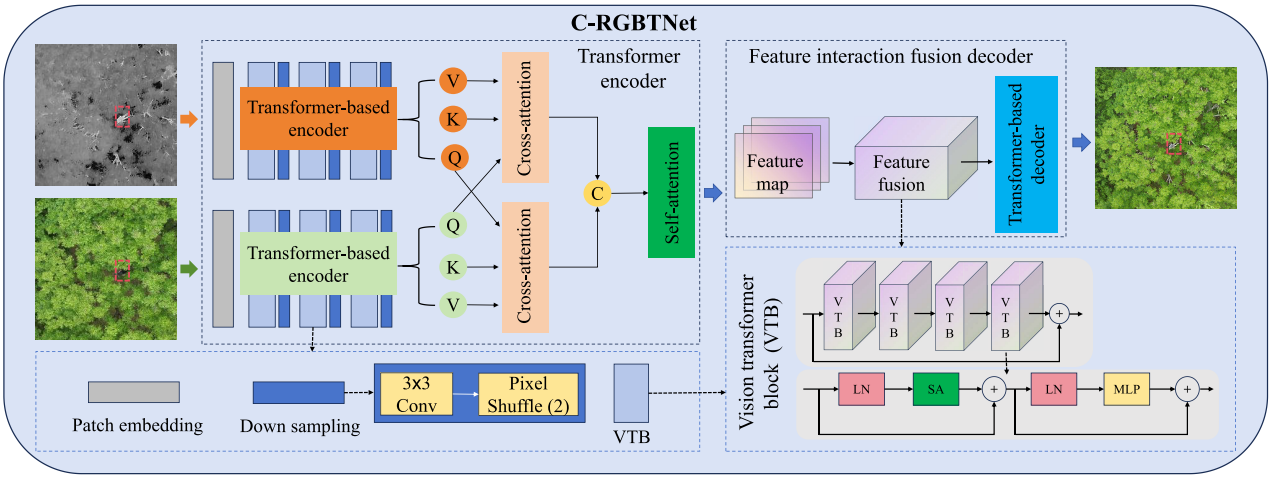


Fig. 4. Structure of transformer-based C-RGBTNet.

features from the input registered thermal and RGB images. Second, a feature interaction fusion module to integrate the modality information. Meanwhile, a transformer decoder reconstructs the fusion image from the integrated features. The output fusion image retains the prominent thermal features of the target and the RGB color features of the occlusions, reducing confusion and enhancing the accuracy of target segmentation.

1) Transformer Encoder: Recent multimodal fusion demonstrates that a transformer can capture global contextual information and align features by leveraging the unique attention mechanism [30], [31]. To extract multimodality image representation, we adopt vision transformer blocks (VTBs) as the base architecture. The architecture of the transformer encoder is illustrated in Fig. 4. The input image is first processed by a patch embedding layer. Each encoder layer consists of a VTB followed by a downsampling module. The downsampling module comprises a 3×3 convolutional layer and a PixelShuffle layer, with a downsampling factor of 2. Given the paired thermal and RGB images as inputs, the encoder process can be described as follows:

$$F_{\text{rgb}} = F_r^E(I_r), \quad F_{\text{thermal}} = F_t^E(I_t) \tag{1}$$

where $I_t \in \mathbb{R}^{H\times W\times 3}$ and $I_r \in \mathbb{R}^{H\times W\times 3}$ denote the thermal and RGB images, respectively. F_t^E and F_r^E represent the thermal and RGB image encoders, respectively. The features extracted

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by the encoder are divided into query (Q), key (K), and value (V), which are then fed into the cross-attention module to align different modalities and facilitate feature interaction. Specifically, it can be described as follows:

$$\begin{aligned}\{Q_r, K_r, V_r\} &= F^r_{qkv}\left(F_{\text{rgb}}\right)\\ \{Q_t, K_t, V_t\} &= F^t_{qkv}\left(F_{\text{thermal}}\right)\end{aligned} \tag{2}$$

where F^r_{qkv} and F^t_{qkv} indicate the transformer functions used to generate Q, K, and V. Simultaneously, we exchange the Q of two modalities for information interaction, that is,

$$\begin{aligned}F^t &= \text{soft}\max\left(\frac{Q_r K_t}{d_k}\right)V_t\\ F^r &= \text{soft}\max\left(\frac{Q_t K_r}{d_k}\right)V_r\end{aligned} \tag{3}$$

where d_k is the scaling factor. At the end of the encoder, the features obtained from the attention module are concatenated to obtain the fused features as follows:

$$F_f = \text{Concat}\left(F^t, F^r\right). \tag{4}$$

2) Feature Interaction Fusion Decoder: As illustrated in Fig. 4, the feature interaction fusion decoder can be divided into the feature fusion module and the transformer-based decoder. Specifically, the feature fusion module consists of a series of VTBs with residual connections, designed to interact with detailed information from both the thermal and RGB images. Given the input fused feature F_f, we apply four VTBs to extract the intermediate global feature F^I. The output is expressed as follows:

$$F^I = F_{\text{VTBs}}\left(F_f\right) + F_f \tag{5}$$

where F_{VTBs} is the vision transformer block. Similar to the ResNet architecture, the multilayer transformer constructs the global feature, while the residual connections aggregate information from different levels. The VTB consists of two-layer normalization (LN), an SA module, and a multilayer perception (MLP) with a Gaussian error linear units (GELUs) nonlinearity function. After the feature fusion module, the fused feature is fed into the transformer encoder to obtain the reconstructed fusion image, expressed as follows:

$$I_{\text{fusion}} = F^D\left(F^I\right) \tag{6}$$

where I_{fusion} is the output fusion image and F^D is the transformer-based decoder.

3) Loss Function: In the training process, we introduce the structure similarity (SSIM), $L1$ gradient, color consistency, and pixel intensity loss. Their functions can be defined as follows:

$$\begin{aligned}L_{\text{SSIM}} &= \left(1 - f_{\text{SSIM}}\left(I_{\text{fusion}} - I_t\right)\right)\\ &\quad + \left(1 - f_{\text{SSIM}}\left(I_{\text{fusion}} - I_r\right)\right)\end{aligned} \tag{7}$$

$$L_{\text{grad}} = \frac{1}{HW}\left\|\nabla I_{\text{fusion}} - \max\left(\nabla I_r, \nabla I_t\right)\right\|_1 \tag{8}$$

$$L_{\text{color}} = \frac{1}{HW}\left\|F_{\text{CbCr}}\left(I_{\text{fusion}}\right) - F_{\text{CbCr}}\left(I_r\right)\right\|_1 \tag{9}$$

$$L_{\text{int}} = \frac{1}{HW}\left\|I_{\text{fusion}} - \max\left(I_r, I_t\right)\right\|_1 \tag{10}$$

where f_{SSIM} is the SSIM operation and F_{CbCr} is the transfer function of RGB to CbCr. In addition, ∇ represents the gradient of the image. Finally, the total loss is defined as follows:

$$L_{\text{total}} = \alpha_1 L_{\text{SSIM}} + \alpha_2 L_{\text{grad}} + \alpha_3 L_{\text{color}} + \alpha_4 L_{\text{int}} \tag{11}$$

where α is the hyperparameters used to adjust the relative weight between the four loss functions. α_1 weights the SSIM loss L_{SSIM}, which measures perceptual differences in luminance, contrast, and structure, promoting high-level structural fidelity. α_2 weights the gradient loss L_{grad}, emphasizing edge sharpness and structural boundaries. α_3 weights the color consistency loss L_{color}, ensuring that the fusion image preserves the chromatic information of the RGB image. α_4 weights the pixel intensity loss L_{int}, encouraging the retention of salient features from both thermal and RGB modalities.

Algorithm 1 HSV-Based White De-Emphasis

Input: RGB images set I

1. Init HSV bounds for white regions: $L = [0, 0, 200]$, $U = [180, 25, 255]$
2. Init predefined color: $C = [128, 128, 128]$
3. **for** image i_R in I
   - Convert i_R (RGB) to i_H (HSV)
   - $M = \text{InRange}(i_H, L, U)$
   - $i_R[\textbf{if } M > 0] = C$
   - Save i_R to O
4. **return** O

Output: Processed RGB images set O

B. HSV-Based White De-Emphasis and Color Segmentation

To address near- or pure-white occlusions in RGB images that could interfere with color segmentation after RGBT fusion, white de-emphasis is applied in the HSV color space. HSV represents colors through three components: hue (color type), saturation (purity), and value (brightness). HSV is particularly advantageous for processing images with rich colors or complex backgrounds [32], [33]. In the HSV color space, white is characterized by high value, low saturation, and unrestricted hue range. In OpenCV [34], white is represented by a hue range of 0–180, a saturation range of 0–25, and a value range of 200–255. The HSV-based white de-emphasis process includes the following steps.

Algorithm 1 processes RGB images by detecting white regions and replacing them with gray. The input is an RGB image set I. First, the algorithm defines the threshold range $[L, U]$ for white in the HSV color space and sets the predefined color C to gray. Next, for each image in I, i_R is converted to image i_H in the HSV color space, and a mask M is created to extract its white regions (InRange). Then, the algorithm searches for white regions in i_R using the mask M and replaces the color values of these regions with C. Finally, the processed i_R is saved in the output image set O.

C-RGBTNet preserves the salient thermal features of the target while maintaining the RGB color features of the occlusions in the fusion image. To extract high-brightness, white-colored target information from the fusion image, we employ HSV-based color segmentation. The HSV-based color segmentation process includes the following steps.

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Algorithm 2 HSV-Based Color Segmentation

Input: Fusion images set I

1. Init $V = [200, 255)$
2. Init HSV bounds for white regions: $L = [0, 0, 200]$, $U = [180, 25, 255]$
3. **for** image i_F in I
   - Convert i_F (RGB) to i_H (HSV)
   - $M = \text{InRange}(i_H, L, U)$
   - $i_F = \text{BitwiseAnd}(i_F, M)$
   - Save i_F to O
4. **return** O

Output: Processed fusion images set O

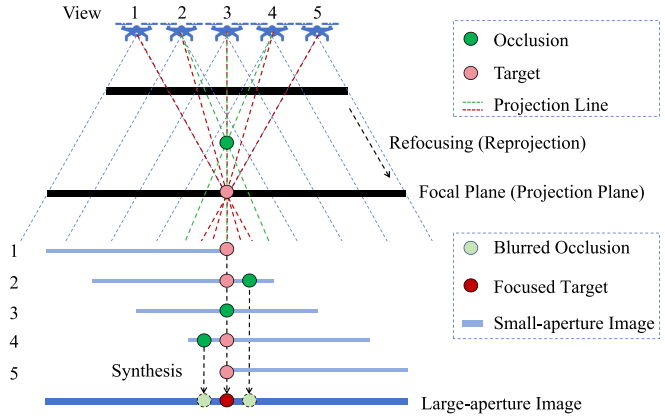


Fig. 5. Principle of occlusion removal target imaging.

Algorithm 2 processes fusion images by segmenting the white regions. The input is a fusion image set I. First, the algorithm defines the brightness range V and the threshold range $[L, U]$ for white in the HSV color space. Next, for each image in I, i_F is converted to image i_H in the HSV color space, and a mask M is created to extract its white regions. Then, the white regions in i_F is retained by performing AND OPERATION (BitwiseAnd) on the pixels in i_F and M. The color values of the remaining regions are replaced with black. Finally, the processed i_F is saved in the output image set O.

C. Image-Rendering-Based Occlusion Removal Target Imaging

As illustrated in Fig. 5 [35], we achieve occlusion removal target imaging by SAI. Fig. 5 illustrates a 2-D plane, where a drone captures downward-view images from five positions. The target and its occlusion form intersection points with the projection lines and the focal plane at each drone position, corresponding to the images captured by the small-aperture camera from different viewpoints. A large-aperture image is obtained by shifting the focal plane to the position of the target (refocusing or reprojection) and synthesizing the refocused images. In the resulting image, the target is sharply focused, while the occlusion is blurred and diffused.We use COLMAP [36], a 3-D reconstruction tool utilizing incremental structure from motion (SfM), to accurately estimate camera poses from multiview images. The pose data, image data, and plane model (serving as the projection plane) are input into OpenGL [37]

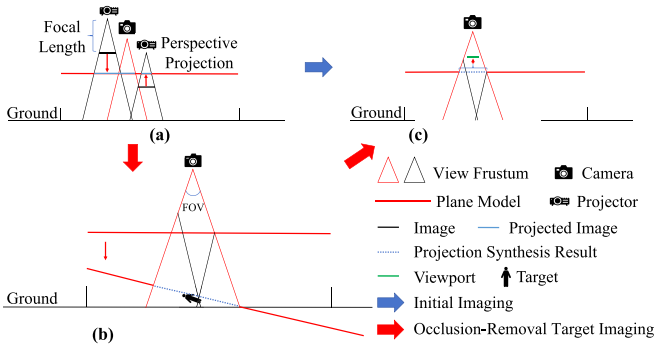


Fig. 6. Image rendering in OpenGL. (a) Image projection. (b) Adjustment of the model. (c) Image rendering.

for rendering, as illustrated in Fig. 6. First, virtual cameras are defined using view transformation matrices, while model transformation matrices position the plane model in front of them. Second, the field of view (FOV) of each frustum is determined using the image resolution and focal length. Perspective projection matrices generate the frustum, positioning the image at the corresponding camera's focal length. Third, a single camera is designated as the rendering camera, with the others functioning as projectors. Images are projected onto the plane model through perspective projection within each frustum, with the initial position of the plane model set at the rendering camera's focal length. Finally, only the plane model within the rendering camera's frustum is rendered. The projections are synthesized and displayed on the screen via viewport transformation. By rotating and translating the plane model to align with the target position, target focusing and occlusion blurring are achieved.

IV. Experiments

We first introduce the home-brew dataset, implementation details, and relevant configurations. Then, we evaluate the effectiveness and advantages of the proposed method through both qualitative and quantitative comparisons. Finally, we conduct ablation experiments on our proposed components.

A. Dataset

Due to the lack of existing downward-view multimodal datasets for occluded scenarios, we built the DMB dataset, comprising six bush scenarios across three seasons and diverse occlusion conditions. As illustrated in Fig. 7, each scenario includes one person lying down in the bush.

1) DMB Dataset: The winter subset includes a slightly occluded scenario [scenario (a)], captured on January 21, 2024. The weather is light snow with a temperature of 5° C, and 176 images are collected. The autumn subset includes a slightly occluded scenario [scenario (c)] and a heavily occluded scenario [scenario (f)], captured on September 20, 2024. Both scenarios are recorded under clear weather, with temperatures of 23° C. A total of 308 images are collected for scenario (c) and 302 images for scenario (f). The summer subset includes a slightly occluded scenario [scenario (b)] and two heavily occluded scenarios [scenarios (d) and (e)],

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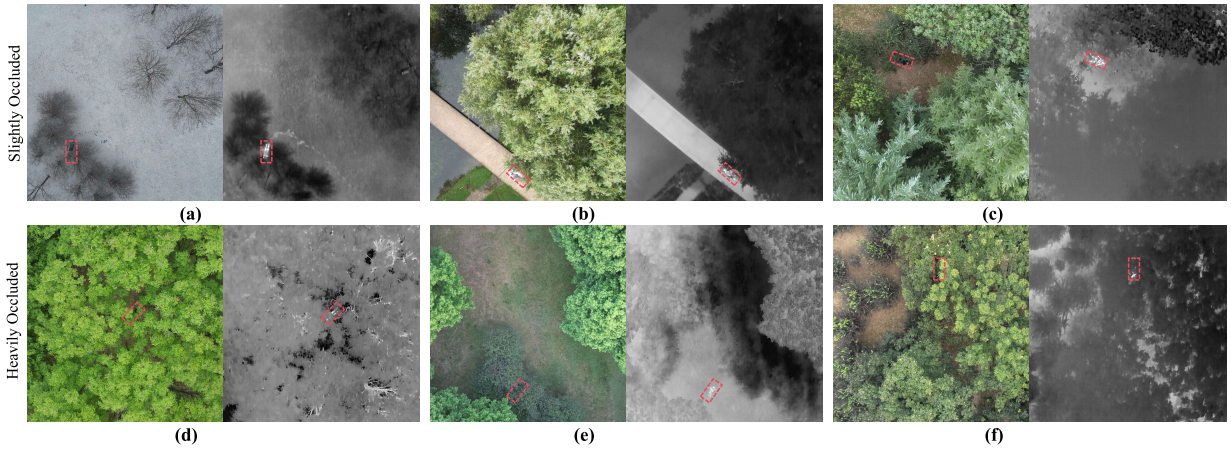


Fig. 7. Representative images from the DMB dataset. "Heavily occluded" refers to cases where the target is almost completely invisible in all captured RGB images, while "slightly occluded" means that the target remains partially visible from most RGB images (and may even be fully visible from certain camera shooting angles). The red dotted boxes indicate the location of the target (a person lying down).

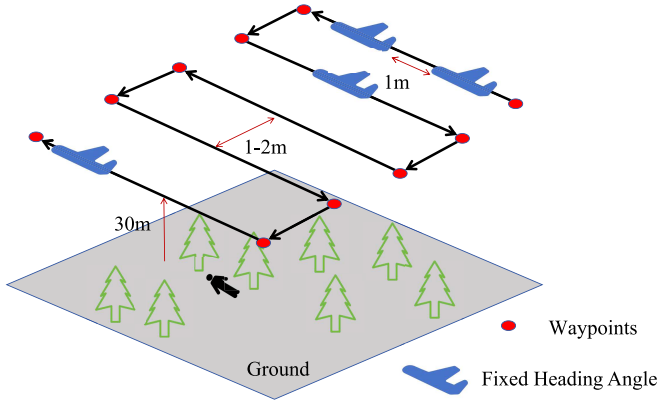


Fig. 8. Schematic of the drone flight mission setup.

captured on July 21, 2024. All scenarios are recorded under clear weather, with temperatures of 36° C. A total of 262 images are collected for scenario (b), 248 for scenario (d), and 254 for scenario (e).

2) Image Acquisition Equipment: A cost-effective DJI Matrice M30T drone is utilized, equipped with a 1280 × 1024 thermal camera (61° FOV, f/1.0, and focus range: 5 m to infinity focus) and a 4000 × 3000 wide-angle camera (84° FOV, f/2.8, and focus range: 1 m to infinity focus).

3) Image Acquisition Conditions: In waypoint flight mode, the drone autonomously follows preset paths and captures images based on programmed tasks. The flight paths are configured over the experiment scenario, as shown in Fig. 8. Key parameters include a flight altitude of 30 m, path spacing of 1–2 m, a shooting interval of 1 m, a flight speed of 1 m/s, a downward-facing gimbal, and a locked heading angle. The entire process is completed in just a few minutes.

4) Data Processing Equipment: Captured images are transferred from the drone to a PC for processing. The PC is equipped with an Intel Core i7-10875H CPU (2.30 GHz), 16-GB RAM, and an RTX 2070 GPU. Image processing is implemented using C++ and Python.

5) Data Processing Method: First, the RGB images are cropped to 1600 × 1200 pixels and the thermal images to 1024 × 1024 pixels. Then, we apply OpenCV for image correction and registration, and COLMAP to estimate the poses of the drone camera by RGB images. Finally, the poses, images obtained from different methods, and a plane model are input into OpenGL for rendering. The entire process takes only a few minutes, with rendering completed in milliseconds.

B. Implementation Details

1) Network Settings: We perform the C-RGBTNet experiments on a computer with two NVIDIA GeForce Titan GPUs, utilizing the PyTorch framework. In the training process, three commonly used RGB and thermal image datasets are LLVIP [38], MSRS [39], and RoadScene [22]. For image augmentation, the original thermal and RGB images are cropped to 96 × 96. In the training process, we use the AdamW optimizer to update the overall network with a learning rate of 0.0001, and the weight decay is 0.005. The batch size is 4, and the training epoch is 130. In (11), hyperparameters are set to 1, 10, 12, and 4, respectively.

2) SOTA Competitors: The existing methods most relevant to our method are the encoder-based models. We compare the proposed C-RGBTNet with representative CNN-based and transformer-based image fusion methods on the DMB dataset. The methods for comparison include U2Fusion [22], CDDFuse [23], and PromptFusion [24].

3) Evaluation Metrics: This section introduces the evaluation metrics used in the experiments, including edge density (ED), image entropy (IE), and signal-to-noise ratio (SNR). These metrics are commonly used in image evaluation and provide a comprehensive view of image quality. For evaluation, the large-aperture images obtained from SAI are converted to grayscale.

4) Edge Density: ED [40] quantifies the proportion of edge pixels in an image, serving as an indicator for its structural richness and complexity. The formula is as follows:

$$\mathrm{ED} = \frac{\mathrm{EP}}{\mathrm{TP}} \tag{12}$$

where EP represents the number of detected edge pixels and TP is the total number of pixels. A higher ED signifies more

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edges in the image, indicating richer image details or more distinct object contours.

5) Image Entropy: IE [41] quantifies the complexity of information within an image by evaluating the distribution of its grayscale levels. The formula is as follows:

$$\mathrm{IE} = -\sum_{i=0}^{L-1} p(i)\log_2 p(i) \tag{13}$$

where L represents the total number of grayscale levels and $p(i)$ represents the probability of grayscale level i. A higher IE reflects greater image complexity with abundant information and details.

6) Signal-to-Noise Ratio: SNR [42] quantifies the ratio between target information and noise, commonly used to assess the clarity and prominence of target features in an image. The formula is as follows:

$$\mathrm{SNR} = 10 \cdot \log_{10}\left(\frac{{\sigma_s}^2}{{\sigma_n}^2}\right) \tag{14}$$

where σ_s represents the variance of the target region, reflecting the extent of grayscale variation within the target, while σ_n represents the variance of noise, indicating the grayscale variation in the background. A higher SNR corresponds to clearer target features and superior image quality.

C. Comparison of Image Fusion Methods

According to Algorithm 1, all RGB images are processed using the HSV-based white de-emphasis method. Fig. 9 illustrates the results of four methods on the DMB dataset. Benefiting from the transformer-based pipeline, which incorporates powerful global context modeling and long-range dependency representation, along with the color consistency loss, C-RGBTNet exhibits three distinctive advantages. First, our results highlight the thermal target more effectively. As shown in the red boxes of Fig. 9, the pixel intensity of the target in our results is markedly stronger than that of U2Fusion, while remaining comparable with CDDFuse and PromptFusion, yet with a more natural appearance. Second, our results preserve the color information of occlusions or objects better than others. As shown in the yellow boxes of Fig. 9, occlusions or objects with higher thermal intensity in our results are also presented with well-preserved RGB texture details. Last, our results provide a more appropriate brightness. As can be observed from Fig. 9, our fusion images maintain a brightness level nearly identical to that of the original RGB image, which prevents false segmentation of occlusions or objects during subsequent HSV-based color segmentation.

1) Comparison of SAFI Results: According to Algorithm 2, the V component in HSV for SAFI is set to 200. Fig. 10 shows the SAFI results of four methods in six experimental scenarios. Compared with U2Fusion, C-RGBTNet preserves target information more effectively, while reducing a substantial number of unwanted regions compared with CDDFuse and PromptFusion. The target pixels of U2Fusion exhibit the lowest intensity, leading to target loss in heavily occluded scenarios. Although the target pixel intensity of CDDFuse is comparable to that of C-RGBTNet, its excessively high brightness causes significant false segmentations in HSV, resulting in many unwanted regions in the SAFI results. In slightly occluded scenarios, the overly bright target of CDDFuse even merges with unwanted regions [scenarios (b) and (c)]. PromptFusion performs comparably to C-RGBTNet in low-temperature scenarios [scenarios (a), (c), and (f)], but it generates more unwanted regions in summer scenarios. C-RGBTNet retains a small number of unwanted regions in certain scenarios [scenarios (b) and (d)], which can be further eliminated by increasing the V, while preserving target information. We will elaborate on this in Section IV-D.

2) Comparison of Evaluation Metric Results: Canny [43] operator is used for edge detection in the experiment. Since the SAFI produces high-contrast images, the threshold is set to 100 and 200. Tables III and IV summarize the evaluation results for each method, highlighting the effectiveness of our C-RGBTNet. PromptFusion achieves comparable metric results to C-RGBTNet in low-temperature scenarios. However, its performance degrades in summer due to the generation of more unwanted regions. CDDFuse yields the worst performance across all metrics due to the presence of numerous unwanted regions. Although U2Fusion achieves better metric results than C-RGBTNet in slightly occluded scenarios, this advantage primarily stems from its fusion images containing limited information and incomplete target details. In contrast, although C-RGBTNet introduces some unwanted regions, they do not compromise the overall visual quality. Moreover, the performance of C-RGBTNet can be further improved by increasing the V component in HSV. For CDDFuse, the excessively high brightness leads to incomplete elimination of unwanted regions even with increased V. In contrast, because U2Fusion already contains almost no unwanted regions, further increases of the V would instead reduce target information and degrade the SNR. For PromptFusion, limited ability to distinguish targets from high-thermal occlusions sets a lower performance ceiling in SAFI than C-RGBTNet, even with increased V.

3) Computational Efficiency Analysis: We evaluate the computational efficiency of all methods using three key metrics: average runtime, number of network parameters, and floating-point operations per second (FLOPs). The evaluation is performed on an NVIDIA GeForce RTX 3090 GPU with ten randomly selected images from the DMB dataset, each with a resolution of 1024× 1024. To ensure accurate runtime measurement, we utilize the official CUDA event functions and include a warm-up phase before the main loop to minimize measurement errors. The results are summarized in Table V. Although C-RGBTNet has more parameters, it achieves lower FLOPs and a significantly shorter average runtime than other methods, which is more efficient in practical applications.

D. Experimental Results

Fig. 11 shows the SAI results across all scenarios. In slightly occluded scenarios, the imaging quality is satisfied ([scenarios (a) and (c)] when the ambient temperature is low. Otherwise, the imaging quality deteriorates [scenario(b)]. In heavily occluded scenarios, the imaged target is dim and fuzzy regardless of the temperature difference [scenarios

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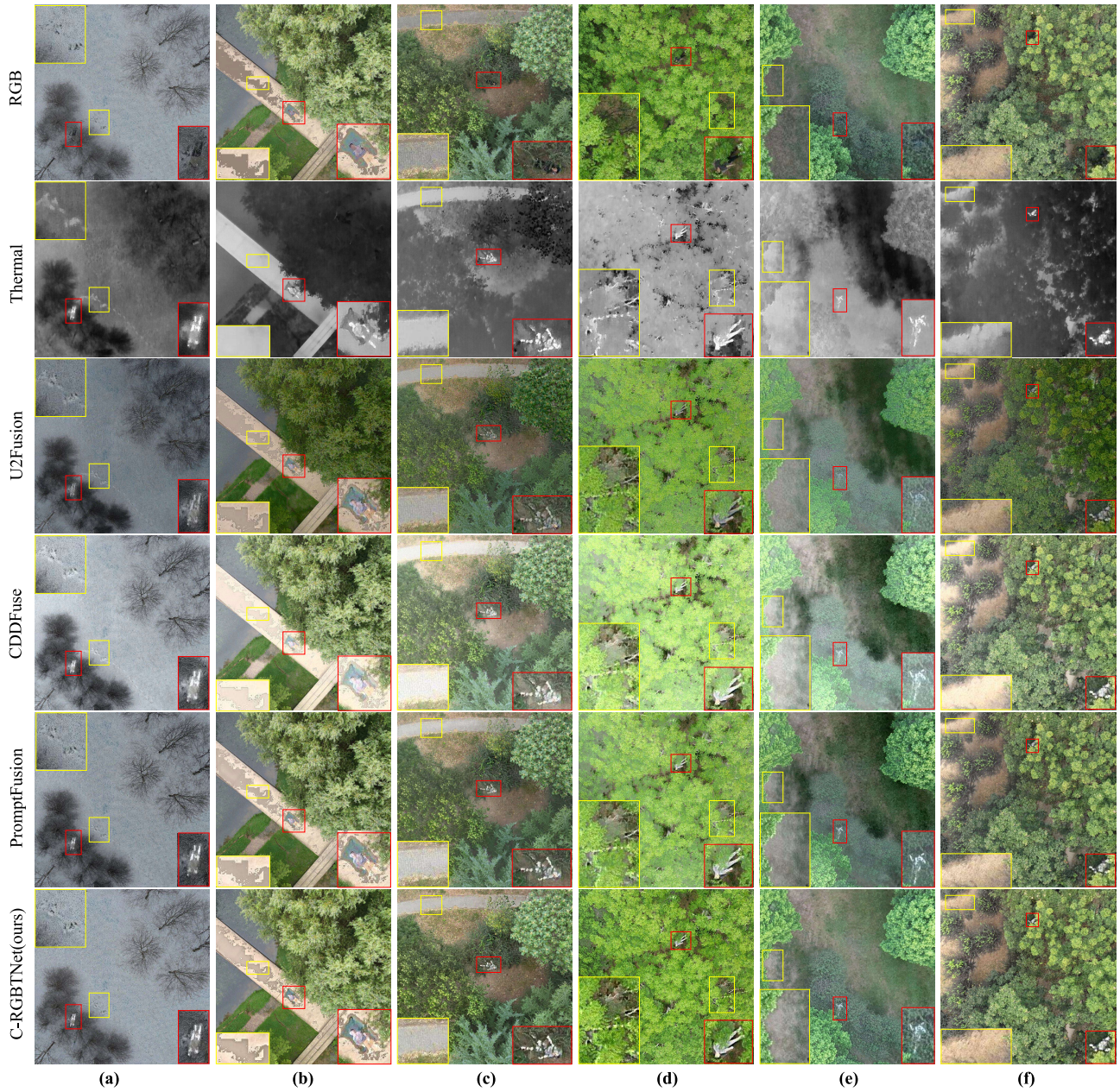


Fig. 9. Comparison of our C-RGBTNet with other image fusion methods.

TABLE III

COMPARISON OF EVALUATION METRICS RESULTS BETWEEN C-RGBTNET AND OTHER METHODS IN SLIGHTLY OCCLUDED SCENARIOS

Methods	(a) Winter			(b) Summer			(c) Autumn		
	ED(%)↓	IE(bits)↓	SNR(dB)↑	ED(%)↓	IE(bits)↓	SNR(dB)↑	ED(%)↓	IE(bits)↓	SNR(dB)↑
U2Fusion	**0.11**	**0.06**	**14.16**	**0.45**	**0.23**	**9.48**	0.20	**0.09**	**17.01**
CDDFuse	9.06	4.18	-8.87	4.90	2.76	-3.04	5.26	1.67	-3.41
PromptFusion	0.40	0.26	10.11	2.30	0.70	2.46	0.16	0.11	15.46
C-RGBTNet (ours)	0.41	0.88	9.87	2.00	0.99	6.34	**0.12**	0.25	15.16

The red number indicates the optimal metric value for its corresponding scenario.

(d)~(f)]. This is primarily due to the occlusions in each image are mixed into the targets during the aperture synthesis. This issue can only be addressed by synthesizing the exposed target information in each image rather than the entire image.

1) SAAI and SAFI: In SAAI, to facilitate the use of a consistent RX threshold τ across different subsets for comparison, we normalized τ and set its range between 0.80 and 0.99. According to Algorithm 2, multiple ranges are established for the V component in HSV for SAFI.

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TABLE IV
COMPARISON OF EVALUATION METRICS RESULTS BETWEEN C-RGBTNET AND OTHER METHODS IN HEAVILY OCCLUDED SCENARIOS

Methods	(d) Summer			(e) Summer			(f) Autumn		
	ED(%)↓	IE(bits)↓	SNR(dB)↑	ED(%)↓	IE(bits)↓	SNR(dB)↑	ED(%)↓	IE(bits)↓	SNR(dB)↑
U2Fusion	0.00	0.02	-	0.00	0.02	-	0.00	0.09	-
CDDFuse	4.92	0.89	0.69	3.19	1.01	-2.02	2.70	1.27	3.49
PromptFusion	1.65	0.35	7.56	0.17	1.01	14.08	**0.14**	**0.06**	12.20
C-RGBTNet (ours)	**0.76**	**0.35**	**10.16**	**0.04**	**0.07**	**15.39**	0.15	0.15	**13.01**

- indicates that the target is lost.

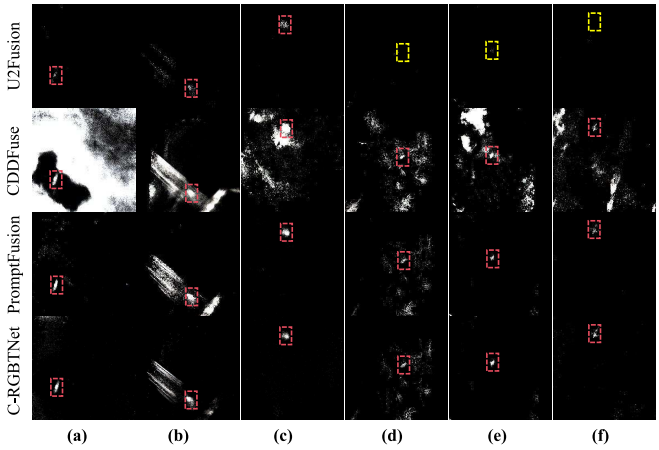


Fig. 10. SAFI results of four methods in six experimental scenarios. The red dotted boxes indicate the target after occlusion removal, while the yellow dotted boxes represent the lost target.

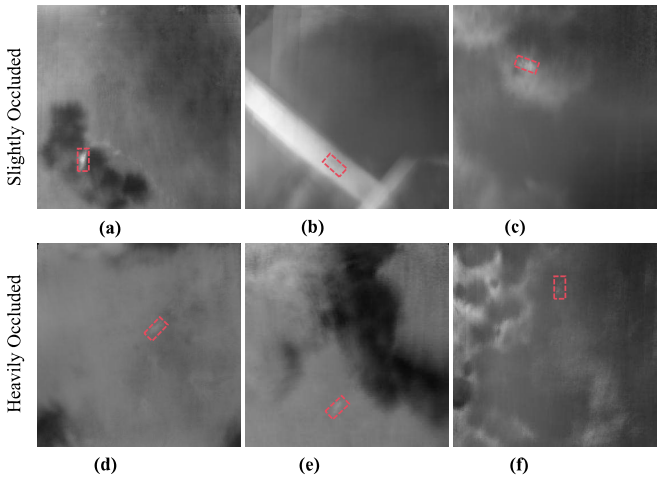


Fig. 11. SAI results of thermal images in six experimental scenarios. The targets are marked by red dotted boxes.

TABLE V
COMPARISON OF COMPUTATIONAL EFFICIENCY ON DMB DATASET

Methods	Time(ms)	FLOPS(G)	Params(M)
U2Fusion	2914	1383	0.659
CDDFuse	1014	1870	1.188
PromptFusion	962.6	447.5	6.727
C-RGBTNet (ours)	161.9	102.4	60.26

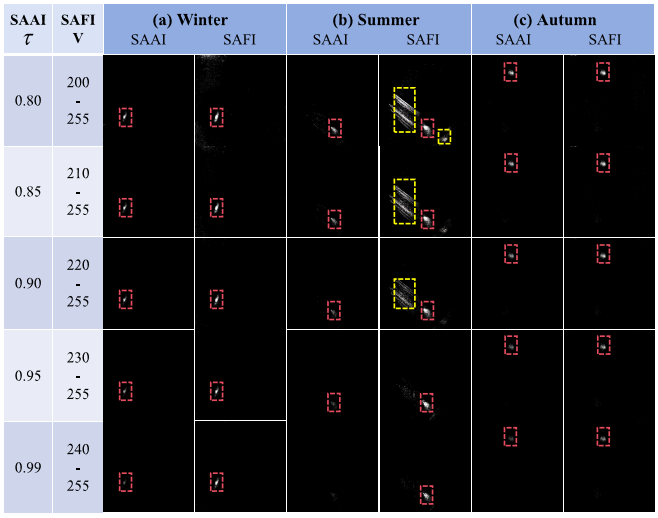


Fig. 12. Results of SAAI and SAFI in slightly occluded scenarios. The red dotted boxes indicate the target after occlusion removal, while the yellow dotted boxes represent the unwanted regions.

The performance of SAAI and SAFI in slightly occluded scenarios is compared in Fig. 12. While SAAI demonstrates overall effectiveness, SAFI achieves superior results in autumn and winter. In summer, SAFI eliminates the unwanted regions caused by thermal cross and preserves the target by setting a higher brightness range. RX processes each pixel from the intensity inconsistency, which does not fully explore the color information of the background and the thermal intensity of the target. In contrast, the HSV-based color segmentation of SAFI skillfully leverages the advantages of both. In conclusion, SAFI focuses on and enhances the target, rendering it visually more distinct.

The performance of SAAI and SAFI in heavily occluded scenarios is compared in Fig. 13. In autumn, SAAI performs well, but in summer, scenario (a) exhibits numerous unwanted regions and target loss, while scenario (b) shows a few unwanted regions. These issues arise from low-thermal occlusions in scenarios (a) and (b). In addition, scenario (a) includes occlusions with thermal intensity similar to the target, as illustrated in Fig. 14. In contrast, SAFI assigns RGB features to both low-thermal occlusions and those with thermal intensity similar to the target. By applying HSV-based color segmentation to remove these occlusions, SAFI demonstrates

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TABLE VI
COMPARISON OF EVALUATION METRICS RESULTS BETWEEN SAAI AND SAFI IN DIFFERENT SCENARIOS

Experimental Scenarios			(a) Winter		(b) Summer		(c) Autumn		(d) Summer		(e) Summer		(f) Autumn	
τ	V	Evaluation Metrics	SAAI	SAFI	SAAI	SAFI	SAAI	SAFI	SAAI	SAFI	SAAI	SAFI	SAAI	SAFI
0.80	200	ED(%)↓	0.15	0.41	0.17	2.00	0.08	0.12	1.50	0.76	0.50	0.04	0.07	0.15
	~	IE(bits)↓	0.08	0.88	0.16	0.99	0.08	0.25	1.31	0.35	0.53	0.07	0.04	0.15
	255	SNR(dB)↑	13.44	9.87	12.82	6.34	15.32	15.16	-	10.16	4.54	15.39	14.78	13.01
0.85	210	ED(%)↓	0.05	0.14	0.13	1.74	0.09	0.08	0.98	0.68	0.41	0.03	0.05	0.07
	~	IE(bits)↓	0.04	0.48	0.10	0.84	0.08	0.13	0.97	0.32	0.40	0.06	0.04	0.11
	255	SNR(dB)↑	18.34	12.80	14.67	7.12	15.31	15.70	-	10.26	5.76	15.50	15.81	13.37
0.90	220	ED(%)↓	0.04	0.10	0.09	1.40	0.08	0.08	0.31	0.55	0.17	**0.02**	0.03	0.07
	~	IE(bits)↓	0.03	0.18	0.07	0.72	0.07	0.09	0.62	0.27	0.19	**0.04**	0.03	0.05
	255	SNR(dB)↑	**19.35**	15.23	13.87	8.28	15.23	**16.61**	-	10.45	9.98	**15.71**	15.15	14.80
0.95	230	ED(%)↓	0.03	0.04	**0.06**	0.45	0.07	0.08	0.07	0.36	0.01	0.00	0.03	0.05
	~	IE(bits)↓	0.03	0.07	**0.06**	0.30	0.06	0.08	0.42	0.21	0.04	0.02	0.03	0.03
	255	SNR(dB)↑	19.03	17.64	13.58	13.59	15.01	15.86	-	**10.86**	-	-	16.49	15.29
0.99	240	ED(%)↓	**0.02**	0.04	0.02	0.24	**0.04**	0.08	0.00	**0.36**	0.01	0.00	0.00	**0.03**
	~	IE(bits)↓	**0.02**	0.04	0.04	0.10	**0.06**	0.07	0.29	**0.19**	0.03	0.02	0.02	**0.03**
	255	SNR(dB)↑	18.70	18.55	-	**15.97**	14.65	15.31	-	9.57	-	-	-	**16.99**
SAI		ED(%)↓	0.02		0		0		0.01		0		0	
		IE(bits)↓	6.70		7.09		6.66		5.57		6.93		6.51	
		SNR(dB)↑	8.32		-		7.97		9.17		5.42		6.10	

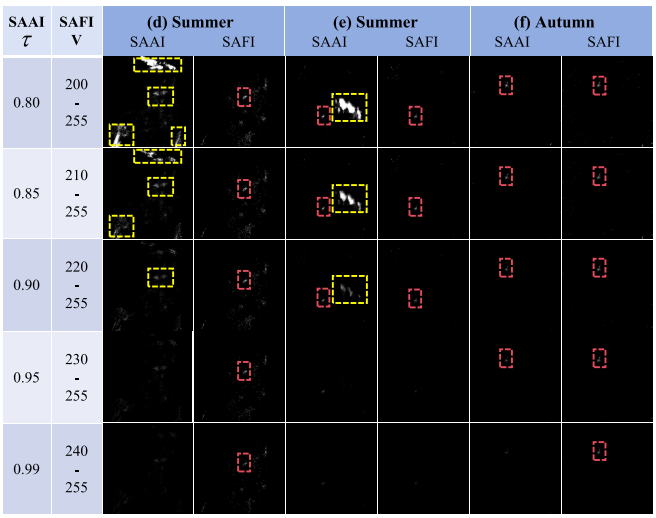


Fig. 13. Results of SAAI and SAFI in heavily occluded scenarios. The red dotted boxes indicate the target after occlusion removal, while the yellow dotted boxes represent the unwanted regions.

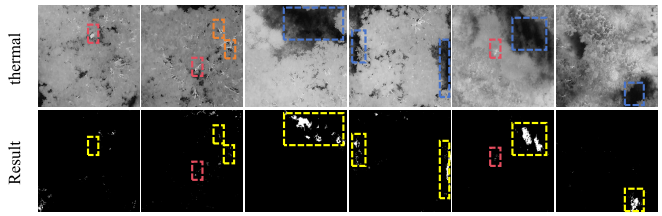


Fig. 14. Illustration of RX detection results for selected thermal images from scenarios (d) and (e). The red dotted boxes indicate the target, the orange for occlusions with similar thermal intensity to the target, the blue for low-thermal occlusions, and the yellow for missed or false detections.

the superior performance across all scenarios, achieving fewer unwanted regions and clearer target imaging.

2) Comparison of Evaluation Metric Results: Since the primary comparison is between SAFI and SAAI, both of which produce high-contrast images, the Canny threshold is set to 100 and 200. Table VI summarizes the evaluation results for two methods, highlighting the effectiveness of the proposed SAFI. Except for the winter subset, where SAAI and SAFI

yield similar metric values, SAFI consistently achieves the highest SNR in other subsets. In heavily occluded scenarios, SAFI exhibits the lowest IE and ED, while in slightly occluded scenarios, SAFI achieves optimal IE and ED nearly identical to SAAI. This demonstrates that SAFI substantially improves the visibility and clarity of occluded targets across diverse conditions. Notably, SAI achieves ED close to 0 across all subsets, which is due to the low contrast of its resulting images. The higher contrast in SAAI and SAFI images renders their Canny threshold unsuitable for SAI.

As the brightness range V and RX threshold τ increase, both IE and ED in SAFI and SAAI decrease due to the progressive loss of image details. However, SNR exhibits different trends, such as an initial increase followed by a decline, a continuous increase, or a steady decrease. For SAFI, these variations result from differences in brightness and quantity between the target and occlusion pixels. When the target exhibits high brightness and the occlusions are relatively dim, increasing V initially eliminates occlusions, thereby improving SNR. However, if V increases further, parts of the target are lost, causing SNR to decline. When the target exhibits extremely high brightness, increasing the V primarily removes occlusions

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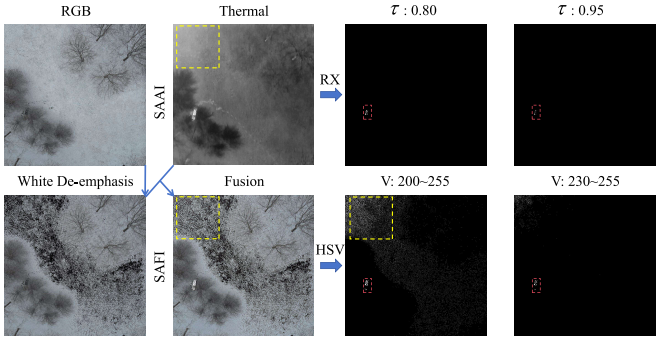


Fig. 15. Image segmentation workflows of SAAI and SAFI in scenario (a). The red dotted boxes indicate the target after segmentation, while the yellow dotted boxes represent unwanted regions after segmentation.

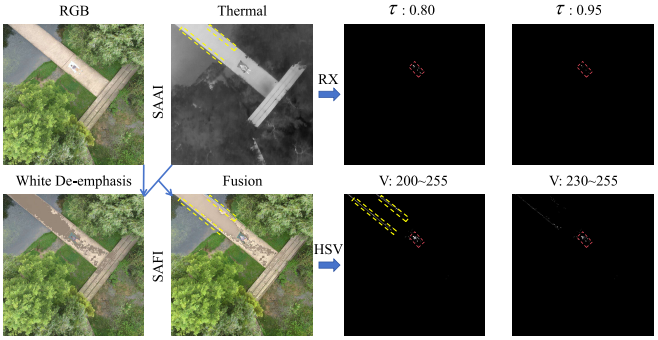


Fig. 16. Image segmentation workflows of SAAI and SAFI in scenario (b). The red dotted boxes indicate the target after segmentation, while the yellow dotted boxes represent unwanted regions after segmentation.

without affecting the target, resulting in a continuously rising SNR. When there are very few occlusion pixels, increasing V mainly causes the loss of low-brightness details in the target, leading to a consistent decrease in SNR. Similarly, for SAAI, variations in SNR arise from differences between target and occlusion pixels, indicating that increasing brightness range V and threshold τ does not always improve results. This highlights the need for multiple V ranges and τ thresholds to achieve optimal performance.

3) Performance Analysis Across Different Seasonal Scenarios: Since SAFI consistently outperforms SAAI in heavily occluded scenarios, the analysis focuses on slightly occluded cases across seasons, where their performance differences are less clear. Specifically, we investigate cases in which the results diverge markedly under two distinct configurations: a threshold τ of 0.80 and 0.95 and a brightness range V of 200~255 and 230~255.

In scenario (a), set in winter, the metric values of SAAI are superior to those of SAFI. However, SAFI retains a more complete and brighter target in the visual. The image segmentation workflows of both methods are illustrated in Fig. 15. In the thermal image, the snow region in the upper left area appears noticeably brighter than its surroundings, resulting in its retention in the fusion image. In scenario (b), set in summer, SAFI attains a higher SNR than SAAI, and both IE and ED are also comparatively higher. The image segmentation workflows of both methods are illustrated in Fig. 16. In the thermal image, the upper bridge part exhibits

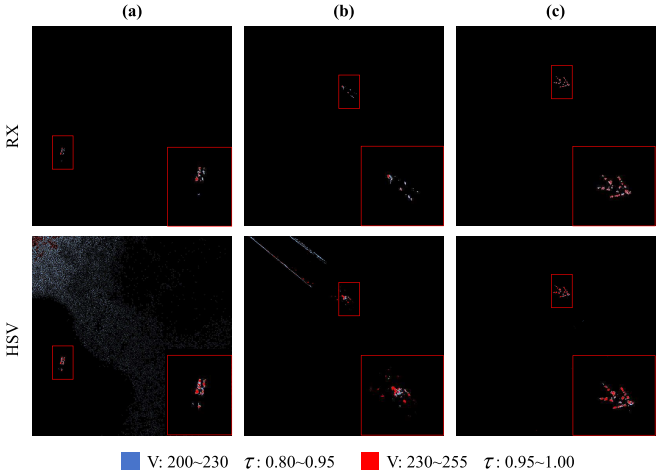


Fig. 17. Pseudocolor images obtained after applying HSV to the fusion images and RX to the thermal images. Red regions correspond to a higher threshold τ and brightness range V than blue regions. The targets are marked by red boxes.

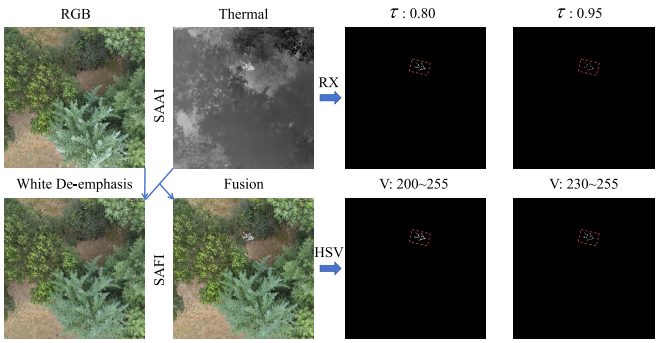


Fig. 18. Image segmentation workflows of SAAI and SAFI in scenario (c). The red dotted boxes indicate the target after segmentation.

a more pronounced brightness contrast with the lake surface. In contrast, the lower bridge part in contact with the grass displays thermal crossover, resulting in a blurred boundary with the land. Consequently, the upper part of the bridge is retained in the fusion image.

In both scenarios (a) and (b), when V is set to 200~255 and the τ is set to 0.80, SAFI exhibits more unwanted regions than SAAI. However, when V is adjusted to 230~255 and τ is adjusted to 0.95, these regions are essentially eliminated, and SAFI retains considerably more target information than SAAI. As shown in Fig. 17, with the RX, the target exhibits distinct color separation, and increasing τ from 0.80 to 0.95 leads to a considerable loss of target information. In contrast, with the HSV, most unwanted regions appear in blue while the target is predominantly red. This allows the adjustment of V from 200~255 to 230~255 to effectively remove unwanted regions with minimal impact on the target. However, a small portion of the unwanted regions appears red, leading to higher ED and IE values for SAFI compared with SAAI due to the increased amount of redundant information. Nonetheless, the SNR remains comparable between SAAI and SAFI because the brightness and completeness of the target in SAFI are enhanced, mitigating the decrease in SNR.

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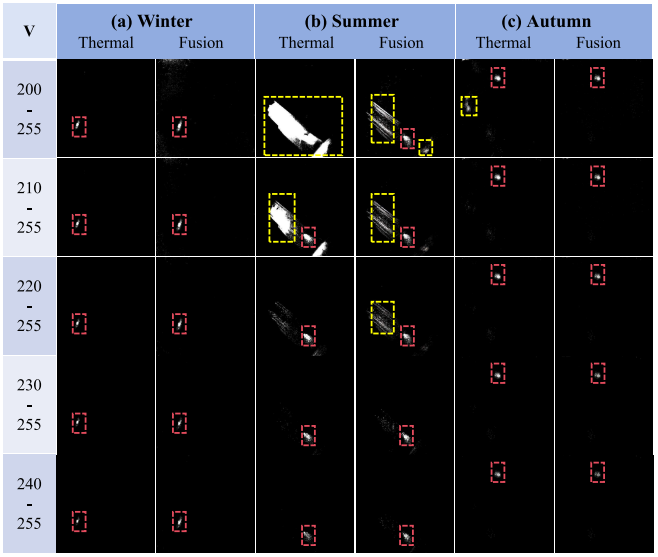


Fig. 19. Results of thermal and fusion images in slightly occluded scenarios. The red dotted boxes indicate the target after occlusion removal, while the yellow dotted boxes represent unwanted regions.

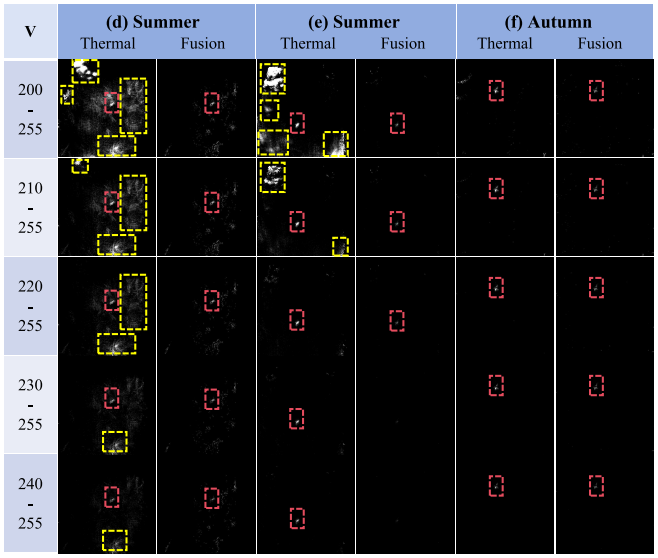


Fig. 20. Results of thermal and fusion images in heavily occluded scenarios. The red dotted boxes indicate the target after occlusion removal, while the yellow dotted boxes represent unwanted regions.

In scenario (c), set in autumn, SAFI achieves a higher SNR than SAAI, with both its IE and ED values also slightly exceeding those of SAAI. The image segmentation workflows of both methods are illustrated in Fig. 18. When V is set to 200~255 and τ is set to 0.80, both methods perform similarly in preserving target information. However, when V is adjusted to 230~255 and τ to 0.95, SAFI retains more target information compared to SAAI. As shown in Fig. 17, with the RX, the target exhibits distinct color separation, and increasing τ from 0.80 to 0.95 leads to a considerable loss of target information. In contrast, with the HSV, the target is predominantly red, indicating that adjusting the brightness range V from 200~255 to 230~255 has minimal effect on the target. Consequently, SAFI obtains a higher SNR than SAAI, while its ED and IE remain nearly identical (with a slight increase due to the additional target information).

The performance of SAAI and SAFI across seasons mainly depends on the temperature contrast between the target and the environment. In spring and autumn, lower ambient temperatures lead to better performance for both methods, whereas higher summer temperatures cause declines in both metrics and visual quality. For SAAI, RX is prone to missing targets because it treats each pixel independently and relies on anomaly scores relative to the global distribution, making it highly sensitive to local brightness variations. Even when the target region is generally bright, localized fluctuations may cause some pixels to fall below the threshold, leading to their omission. In contrast, for SAFI, HSV relies on overall brightness; as long as the brightness of the target falls within the set range, the entire region is preserved, and minor variations rarely cause pixel-level omissions.

4) Ablation Experiments: We replace the fusion image with the thermal image, apply HSV color segmentation directly, and use the results for SAI. Figs. 19 and 20 present a comparison of the experimental results for thermal and fusion images under slightly and heavily occluded scenarios, respectively. Both types of images perform well in winter and autumn. In summer, higher ambient temperatures cause occlusions with strong thermal intensity to generate many unwanted regions in thermal images. By setting a higher brightness range, unwanted regions are reduced in scenarios (b) and (e). However, they persist in scenario (d) due to the near-identical thermal intensity between certain occlusions and the target. The fusion image emphasizes target thermal features while assigning RGB colors to similar thermal occlusions, greatly reducing unwanted regions and enhancing image quality. Furthermore, unlike the thermal image, the fusion image typically provides optimal visual results directly, eliminating the need for repeated brightness adjustments.

Table VII summarizes the evaluation results for both types of images, highlighting the effectiveness of the fusion image. Except for the winter subset, where the results are comparable, the fusion image outperforms the thermal images in overall metrics across other subsets. The thermal image exhibits frequent unwanted regions, primarily in smooth areas. In contrast, the fusion image shows fewer unwanted regions, mainly appearing as short lines or isolated points. Consequently, the improvement in ED for the fusion image is minimal, but it consistently achieves the lowest IE. As shown in Fig. 21, in summer scenarios, the fusion image assigns RGB color features to occlusions with thermal intensity similar to that of the target. Compared with the thermal image, the fusion image reduces the brightness of occlusions and significantly diminishes their impact on the target during HSV-based color segmentation. However, target preservation in the fusion image during summer is inferior to that observed in the thermal image. This discrepancy is primarily attributed to the fusion process, wherein regions of the target exhibiting lower thermal intensity are also assigned RGB color features. Specifically, in scenario (d), the central area of the target displays lower

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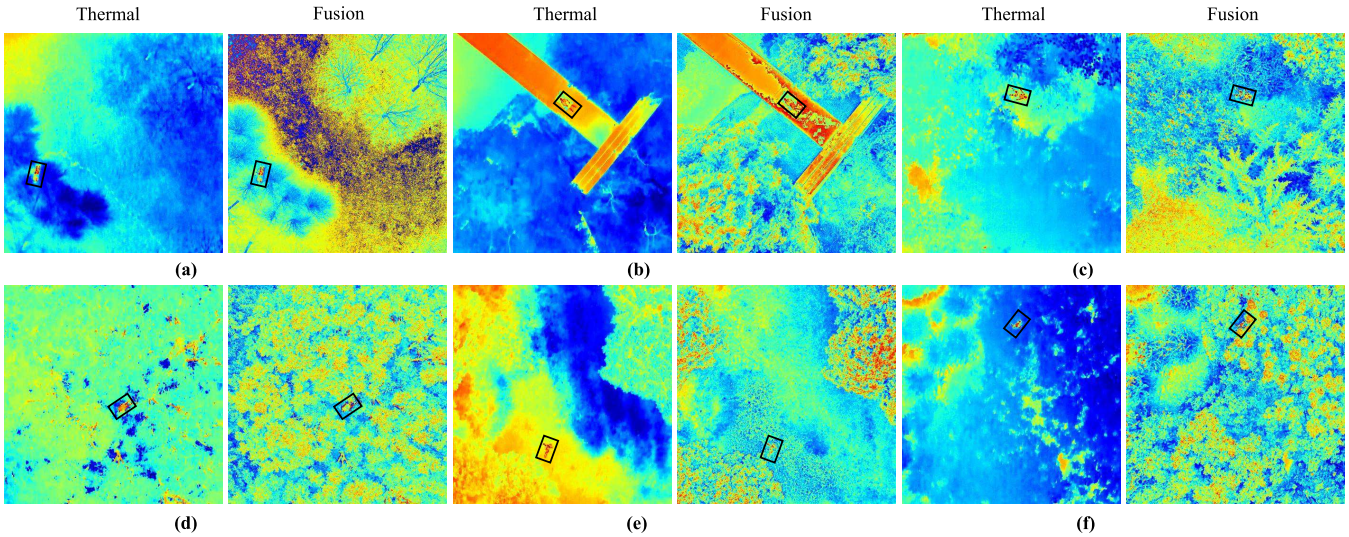


Fig. 21. Heatmaps of representative fusion images and thermal images in all scenarios. The targets are marked by black boxes.

TABLE VII
COMPARISON OF EVALUATION METRICS RESULTS BETWEEN THERMAL AND FUSION IMAGES IN DIFFERENT SCENARIOS

Experimental Scenarios		(a) Winter		(b) Summer		(c) Autumn		(d) Summer		(e) Summer		(f) Autumn	
V	Evaluation Metrics	Thermal	Fusion	Thermal	Fusion	Thermal	Fusion	Thermal	Fusion	Thermal	Fusion	Thermal	Fusion
200	ED(%)↓	0.07	0.41	1.01	2.00	0.31	0.12	3.72	0.76	2.05	0.04	0.21	0.15
∼	IE(bits)↓	0.05	0.88	0.98	0.99	0.37	0.25	3.75	0.35	2.16	0.07	0.17	0.15
255	SNR(dB)↑	18.15	9.87	-	6.34	13.49	15.16	6.60	10.16	6.69	15.39	11.24	13.01
210	ED(%)↓	0.08	0.14	1.67	1.74	0.14	0.08	1.80	0.68	0.91	0.03	0.10	0.07
∼	IE(bits)↓	0.05	0.48	1.01	0.84	0.16	0.13	2.58	0.32	1.03	0.06	0.08	0.11
255	SNR(dB)↑	18.50	12.80	5.02	7.12	15.30	15.70	7.85	10.26	9.30	15.50	14.24	13.37
220	ED(%)↓	0.07	0.10	0.95	1.40	0.11	0.08	1.01	0.55	0.20	**0.02**	0.07	0.07
∼	IE(bits)↓	0.04	0.18	0.11	0.72	0.58	0.09	1.95	0.27	0.31	**0.04**	0.04	0.05
255	SNR(dB)↑	18.43	15.23	10.66	8.28	15.85	**16.61**	8.41	10.45	14.37	15.71	**17.15**	14.80
230	ED(%)↓	0.06	0.04	0.22	0.45	0.09	0.08	0.56	0.36	0.08	0.00	0.06	0.05
∼	IE(bits)↓	0.04	0.07	0.20	0.30	0.09	0.08	1.48	0.21	0.13	0.02	0.03	0.03
255	SNR(dB)↑	18.45	17.64	15.21	13.59	15.66	15.86	8.97	**10.86**	16.26	-	17.10	15.29
240	ED(%)↓	0.04	**0.04**	**0.19**	0.24	0.09	**0.08**	**0.31**	0.36	0.05	0.00	0.06	**0.03**
∼	IE(bits)↓	0.04	**0.04**	0.11	**0.10**	0.08	**0.07**	1.05	**0.19**	0.07	0.02	0.03	**0.03**
255	SNR(dB)↑	**18.68**	18.55	15.03	**15.97**	15.96	15.31	8.09	9.57	**16.67**	-	17.04	16.99

thermal intensity due to obstruction by clothing, which leads to the incorporation of the color information from clothing into the target. In contrast, in the winter and autumn scenarios, the lower ambient temperatures ensure that the target preservation in the fusion image is comparable to that in the thermal image. Although the thermal image retains more target information than the fusion image, its excessive unwanted regions lead to less stable SNR.

V. CONCLUSION

In this article, we focus on enhancing the imaging quality of occluded targets in drone-captured downward-view images to improve detection accuracy in bush search and rescue. We propose SAFI, an ASAI method that integrates a Transformer-based C-RGBTNet with SAI. C-RGBTNet fuses thermal and RGB information to produce high-quality images, emphasizing target thermal features while incorporating RGB cues from occlusions to prevent confusion. HSV-based color segmentation is then applied to extract target thermal features, enabling SAI to achieve high-quality occlusion-free imaging. However, SAFI faces limitations in preserving target information due to thermal feature loss during fusion, and its performance in summer requires improvement. Future work will focus on optimizing the fusion network and enhancing pose estimation to reduce computation and improve accuracy.

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